

Kenya Smart Agriculture

& Market Intelligence Platform

Using NASA satellite weather data to predict food security risk, forecast food prices, and recommend crops to farmers across all 47 counties in Kenya.

47

Counties

409,811

NASA Records

226

WFP Markets

5

ML Models



The Problem



Response is Too Late

Food security phases are assessed manually every 3 months. By the time a county is declared in crisis, the situation has been deteriorating for months. Emergency aid arrives after the crisis has peaked.



No Early Warning System

Rainfall deficits cause food price spikes 2–3 months later. This pattern is predictable — but no automated tool connects NASA weather data to market prices. Farmers get blindsided.



Data Exists But Is Fragmented

NASA has 24 years of weather data. FEWS NET has IPC classifications. KNBS has monthly price reports. WFP has 226 market price records. None of it had ever been merged and used predictively.



The Solution — Four Integrated ML Modules



Food Security Classifier

Which counties are heading into food crisis?

XGBoost

F1 = 0.738

✓ **TARGET MET**



Price Forecasting

How will food prices change in 8 months?

Prophet

MAPE = 0.81%

✓ **TARGET MET**



Crop & Market Recommendation

What should I plant and where to sell?

NASA + WFP

226 markets

✓ **TARGET MET**



NLP Sentiment Analysis

What are farmers reading and feeling?

VADER

47.3% positive

✓ **TARGET MET**



Raw Data — 5 Datasets, All Non-Curated



NASA POWER Weather

409,811 rows

2000–2023

Daily weather for all 47 counties — temperature, rainfall, humidity, solar radiation. The primary ML signal.



FEWS NET IPC

640 rows

Mar 2026

Ground truth labels for the classifier — food security phases 1 to 3 at sub-county level across Kenya.



KNBS CPI Reports

37 raw PDFs

2020–2025

Kenya National Bureau of Statistics monthly price reports. Extracted using regex from unstructured PDF text.



WFP Food Prices

19,005 rows

2006–2026

World Food Programme market monitoring across 226 Kenya markets. Powers the crop price recommendations.



Kenya Agri News

300 articles

2025–2026

Scraped from Kenya News Agency. Used for VADER sentiment analysis and TF-IDF topic modelling.



Model 1 — Food Security Risk Map

12 Counties

PHASE 3 — CRISIS

Turkana, Garissa, Wajir,
Mandera, Marsabit + 7 more

9 Counties

PHASE 2 — STRESSED

Baringo, Kilifi, Kajiado,
Embu, Samburu + 4 more

26 Counties

PHASE 1 — MINIMAL

Nairobi, Nakuru, Kiambu,
Nakuru, Kisumu + 21 more



Key Insight: NASA SPI-3 (Standardised Precipitation Index) is the strongest predictor of IPC food security phase.
All 12 Crisis counties are in Kenya's northern arid and semi-arid zones — this is structural, not seasonal.

Model Performance

Weighted F1 Score

0.738

Target: > 0.70

Phase 3 Crisis F1

0.81

Best at detecting worst cases

5-Fold CV Mean F1

0.732

No overfitting confirmed

Primary Feature

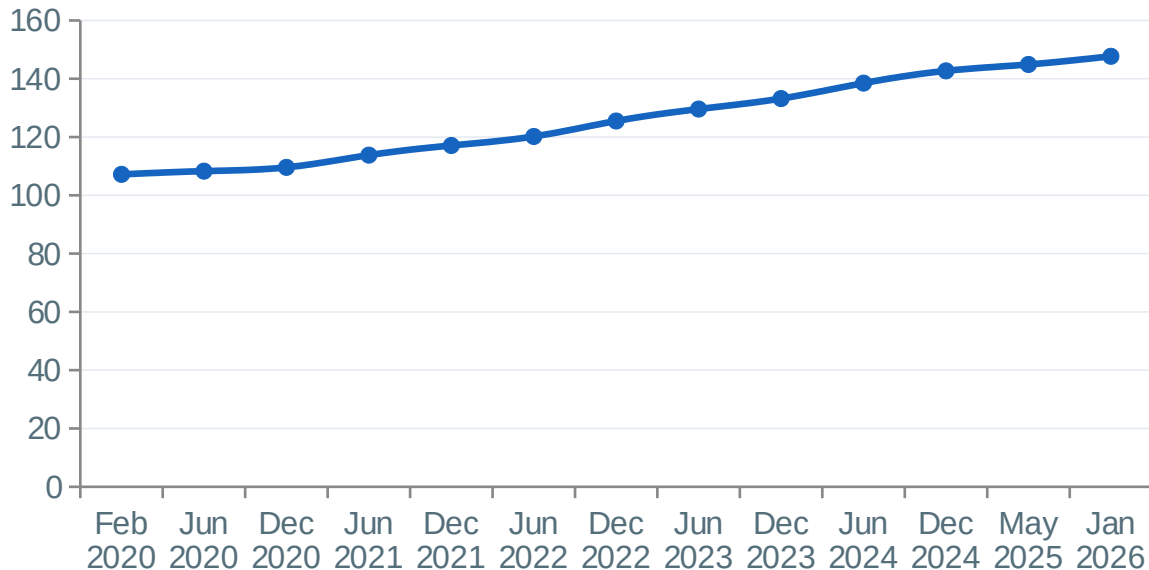
SPI-3

NASA drought index



Model 2 — Kenya Food Price Forecast

Kenya Overall CPI — Historical + Forecast (Feb 2020–Jan 2026)



+35%

Price rise
2020→2025

0.81%

MAPE
(Prophet model)

147.67

Forecast CPI
Jan 2026



Rainfall deficits cause food price spikes 2–3 months later. When NASA SPI-3 drops below -0.5, food prices spike. Pre-positioning food stocks at this signal prevents the worst price shocks for farmers.



Module 3 — Farmer Crop & Market Recommendation

A farmer in any of Kenya's 47 counties selects their county and season. The platform scores 10 crops against NASA weather data and shows current WFP market prices per kg.



Turkana County

Arid • 94% drought • Phase 3 Crisis



Cassava

Score: 100% • N/A



Cowpeas

Score: 94% • KES 138/kg



Sorghum

Score: 90% • KES 76/kg

All drought-tolerant crops selected automatically
based on 24 years of NASA rainfall data



Nakuru County

Highland • Adequate rainfall • Phase 1 Minimal



Potatoes (Irish)

Score: 87% • KES 75/kg



Beans

Score: 87% • KES 130/kg



Kale (Sukuma)

Score: 87% • KES 91/kg

Cool highland crops matched to temperature
and rainfall — different county, different crops

Module 4 — Agricultural News Sentiment Analysis

142

Positive

47.3%

128

Neutral

42.7%

30

Negative

10.0%

Government Programmes

Score: 0.42 • Most positive topic

Soil & Yield Improvement

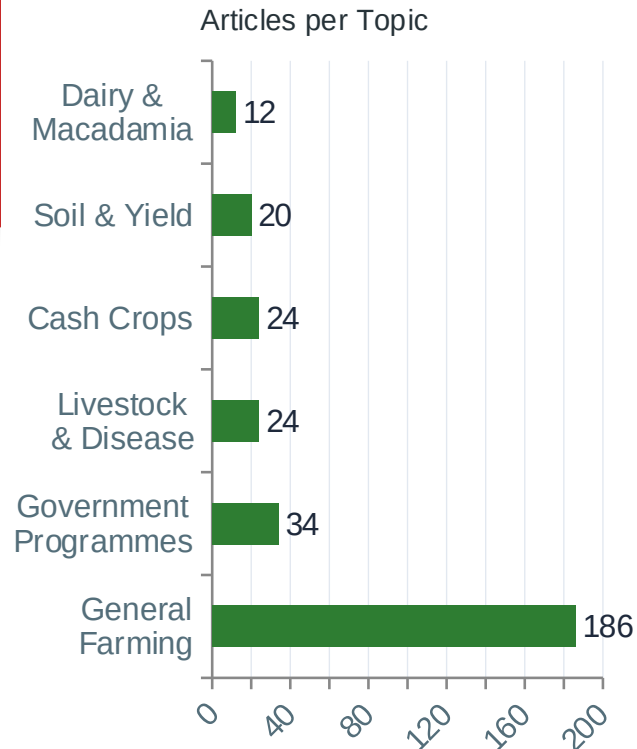
Score: 0.32 • Second most positive

Dairy & Macadamia

Score: 0.11 • Most negative — macadamia export ban

Mean VADER Score

Score: 0.174 • Net positive overall





Key Findings & Recommendations

12 counties in Phase 3 Crisis

Prioritise Turkana, Garissa, Wajir, Mandera, Marsabit for immediate food aid and cash transfer interventions.

NASA SPI-3 predicts food crises months in advance

Deploy automated monthly drought alerts to county agricultural officers when SPI-3 drops below -1.0.

2–3 month lag between drought and price spikes

WFP and NGOs should pre-position food stocks when SPI-3 drops below -0.5 — before prices spike.

Kenya CPI rose 35% from 2020 to 2025

Prophet forecasts CPI reaching 148 by January 2026 — prepare for continued food price pressure.

Negative news sentiment precedes formal IPC assessments

Integrate news sentiment monitoring as an early warning signal to trigger faster field verification.



Live Platform — Try It Now



Streamlit App

kenya-smart-agriculture.streamlit.app



Home — model results + key findings



Food Security Map — all 47 counties



Price Forecast — CPI + 8-month outlook



Crop Recommendation — by county + season



News Sentiment — 300 headlines analysed



Tableau Dashboard

public.tableau.com — Eve Michelle



Food Security Map — Kenya counties



CPI Price Trend — 2020 to 2026



ML Model Performance Results



News Topics & Sentiment



County Mentions in Headlines



Kenya Smart Agriculture

& Market Intelligence Platform

Built by Eve Otieno — Data Scientist, Nairobi, Kenya



kenya-smart-agriculture.streamlit.app



public.tableau.com — Eve Michelle



github.com/EveMichelle/kenya-smart-agriculture